Open Problems

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1 Token Sliding Graphs
2 Distance Recoloring

(Combinatorial) Reconfiguration

studies problems in the following setting:

For any source problem $\mathcal{P}$ (e.g., *Satisfiability*)

- Each *configuration* is a feasible solution of $\mathcal{P}$ (e.g., *satisfying truth assignment*)
  - In general, *the set of all configurations is huge and not part of the input*
- For two configurations A, B, B is *adjacent* to A if it can be obtained from A by applying a given (reconfiguration) rule (e.g., *flipping one bit*) exactly once

Typical questions are:

- **Algorithmic:** *Is there a sequence* of adjacent configurations between two given ones? If so, *is this shortest*? *Can we bound the length* (= the number of times the rule is applied) of such a sequence, if it exists?, and so on
- **Graph-Theoretic:** From the above setting, a *(reconfiguration) graph* can be naturally defined. *Which graph class* does this graph belong to? *Which property* does this graph satisfy? Is this *closed under a graph operation*?, and so on

Token Sliding Graphs: Definition

Token Sliding Graphs ($\mathsf{TS}(G)$ and $\mathsf{TS}_k(G)$)

- **Vertex:** (Size-k) *stable set* of G ($1 \leq k \leq \alpha(G)$)
- **Edge:** Defined via *Token Sliding*

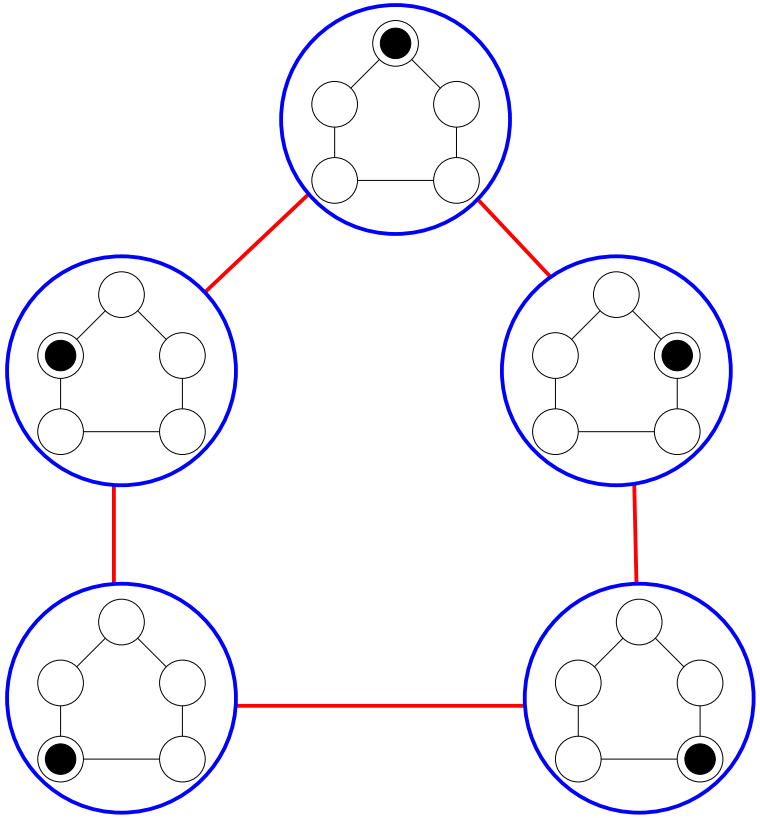

Figure: $\mathsf{TS}_1(C_5)$

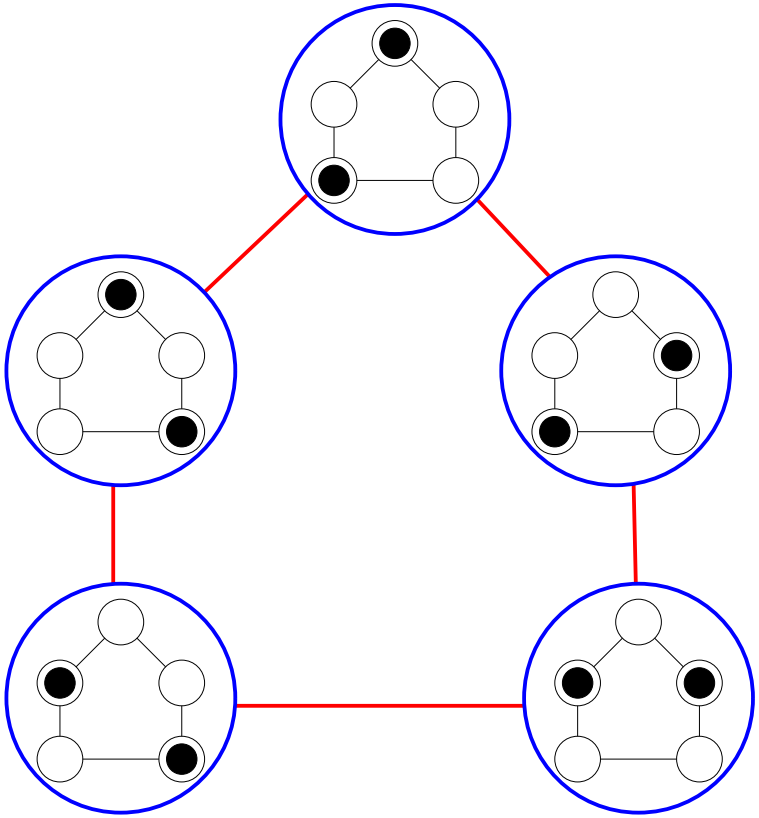

Figure: $\mathsf{TS}_2(C_5)$

Token Sliding Graphs: Motivation

- *Algorithmic questions* have been *well-studied* ([Nishimura 2018], Sections 4 and 5), while *graph-theoretic questions* have *not*
- Similar graphs that have been well-studied in the literature
  - *Fibonacci cube*, first introduced in [Hsu 1993] (see the survey [Klavžar 2013]) [Reconfiguration graph of independent sets on paths under Token Addition/Removal]
  - *Simplex graphs*, first introduced in [Bandelt and De Vel 1989] [Reconfiguration graph of cliques under Token Addition/Removal]
  - *Token graphs*, first studied in [Alavi, Behzad, Erdős, and Lick 1991] (see the survey [Monroy et al. 2012]) [Reconfiguration graph of vertex subsets under Token Sliding]
  - *Matching graphs*, first introduced in [Eroh and Schultz 1998] [Reconfiguration graph of (maximum) matchings under Token Addition/Removal]
  - *Reconfiguration graphs of dominating sets and vertex-colorings*, see the survey [Mynhardt and Nasserasr 2019]

Token Sliding Graphs: Open Problem

We initiate the study of $\mathsf{TS}(G)$ and $\mathsf{TS}_k(G)$ from the graph-theoretic viewpoint

- David Avis and Duc A. Hoang (2023b). "On Reconfiguration Graph of Independent Sets under Token Sliding". In: *Graphs and Combinatorics* 39.3. (article 59). DOI: 10.1007/s00373-023-02644-w. arXiv: 2203.16861
- David Avis and Duc A. Hoang (2023a). "A Note On Acyclic Token Sliding Reconfiguration Graphs of Independent Sets". In: *arXiv preprint*. arXiv: 2301.00317

Some open questions from our research

1. **Given a graph G. What are the necessary and sufficient conditions for G such that $\mathsf{TS}_k(G)$ is a forest?**
   - We partially answer this question in [Avis and Hoang 2023a] when *G is a forest* for $k = 2, 3$ and give a conjecture for $k \geq 4$
   - The case *G is not a forest* requires more insightful ideas
2. **Given a graph G. What are the necessary and sufficient conditions for G to be a TS_k-graph (of some graph H)?**
   - We answer this question in [Avis and Hoang 2023b] for the case G is *a path, a cycle, a complete graph, a complete bipartite graph, or a connected split graph*
   - What about *other graph classes*?

Distance Recoloring: Definition

- A *proper k-coloring ($k \geq 2$)* of a graph G is a function α that assigns a color in $\{1, \ldots, k\}$ to each vertex of G such that *no two adjacent vertices share the same color*
- A *(d, k)-coloring ($d \geq 1$ and $k \geq d + 1$)* of a graph G is a function α that assigns a color in $\{1, \ldots, k\}$ to each vertex of G such that *no two vertices within distance d from each other share the same color*
  - In particular, a *$(1, k)$-coloring* is *a proper k-coloring*
  - The (d, k)-coloring concept was *first introduced* in [F. Kramer and H. Kramer 1969b]; [F. Kramer and H. Kramer 1969a]
- In 2007, Sharp [Sharp 2007] proved that: *Deciding if a graph G has a (d, k)-coloring ($d \geq 2$) is* NP*-complete for $k > \lfloor 3d/2 \rfloor$ and in* P *for $k \leq \lfloor 3d/2 \rfloor$*
  - Before this, the problem is known to be NP*-complete for $d \geq 2$ and large k*
- See the survey [F. Kramer and H. Kramer 2008]

Distance Recoloring: Motivation

(d, k)-Coloring Reconfiguration
Input: (G, α, β), where α and β are two (d, k)-colorings of G
Question: Does there exist a sequence of (d, k)-colorings between α and β such that each intermediate (d, k)-coloring is obtained from its predecessor by recoloring exactly one vertex?

- *Reconfiguration of k-colorings ($d = 1$) has been well-studied* from both algorithmic and graph-theoretic viewpoints [Nishimura 2018]; [Mynhardt and Nasserasr 2019]
  - *in* P when $k \leq 3$
  - PSPACE-*complete* on *bipartite graphs* when $k \geq 4$
  - PSPACE-*complete* on *planar graphs* when $4 \leq k \leq 6$, and *in* P for $k \geq 7$
  - PSPACE-*complete* on *bipartite planar graphs* when $k = 4$, and *in* P for $k \geq 5$
- On the other hand, *reconfiguration of (d, k)-colorings ($d \geq 2$) has not yet been studied*

Distance Recoloring: Open Problem

We initiate the study of reconfiguring (d, k)-colorings from the algorithmic viewpoint

- Niranka Banerjee, Chistian Engels, and Duc A. Hoang (2024). "Distance Recoloring". In: *arXiv preprint*. arXiv: `2402.12705`

Some open questions from our research

1. **What is the complexity of reconfiguring (d, k)-colorings on trees?**
   - We prove in [Banerjee, Engels, and Hoang 2024] that the problem remains PSPACE-*complete* even on *bipartite, planar, and 2-degenerate graphs* for $d \geq 2$ *and* $k = \Omega(d^2)$
   - What about 1-*degenerate graphs* (≡ forest)?
     - We partially answer this question by designing a $O(n^2)$-*time algorithm on paths*
2. **What about other graph classes?**
   - We also prove in [Banerjee, Engels, and Hoang 2024] that the problem remains PSPACE-*complete* on *split graphs* for $d = 2$ *and large* k, while in P for $d \geq 3$ *and* $k \geq d + 1$
   - What about *other graph classes*?

References

- Banerjee, Niranka, Chistian Engels, and Duc A. Hoang (2024). "Distance Recoloring". In: *arXiv preprint*. arXiv: 2402.12705.
- Avis, David and Duc A. Hoang (2023a). "A Note On Acyclic Token Sliding Reconfiguration Graphs of Independent Sets". In: *arXiv preprint*. arXiv: 2301.00317.
- Avis, David and Duc A. Hoang (2023b). "On Reconfiguration Graph of Independent Sets under Token Sliding". In: *Graphs and Combinatorics* 39.3. (article 59). DOI: 10.1007/s00373-023-02644-w. arXiv: 2203.16861.
- Mynhardt, C.M. and S. Nasserasr (2019). "Reconfiguration of Colourings and Dominating Sets in Graphs". In: *50 years of Combinatorics, Graph Theory, and Computing*. Ed. by Fan Chung, Ron Graham, Frederick Hoffman, Ronald C. Mullin, Leslie Hogben, and Douglas B. West. 1st. CRC Press, pp. 171–191. DOI: 10.1201/9780429280092-10.

References

- Nishimura, Naomi (2018). “Introduction to Reconfiguration”. In: *Algorithms* 11.4, p. 52. DOI: 10.3390/a11040052.
- Klavžar, Sandi (2013). “Structure of Fibonacci cubes: a survey”. In: *Journal of Combinatorial Optimization* 25.4, pp. 505–522. DOI: 10.1007/s10878-011-9433-z.
- Monroy, Ruy Fabila, David Flores-Peñaloza, Clemens Huemer, Ferran Hurtado, Jorge Urrutia, and David R. Wood (2012). “Token Graphs”. In: *Graphs and Combinatorics* 28.3, pp. 365–380. DOI: 10.1007/s00373-011-1055-9.
- Kramer, Florica and Horst Kramer (2008). “A Survey on the Distance-Colouring of Graphs”. In: *Discrete mathematics* 308.2-3, pp. 422–426. DOI: 10.1016/j.disc.2006.11.059.
- Sharp, Alexa (2007). “Distance Coloring”. In: *Proceedings of ESA 2007*. Vol. 4698. Lecture Notes in Computer Science. Springer, pp. 510–521. DOI: 10.1007/978-3-540-75520-3_46.

References

- Eroh, Linda and Michelle Schultz (1998). “Matching graphs”. In: *Journal of Graph Theory* 29.2, pp. 73–86. DOI: 10.1002/(SICI)1097-0118(199810)29:2<73::AID-JGT3>3.0.CO;2-9.
- Hsu, Wen-Jing (1993). “Fibonacci cubes-a new interconnection topology”. In: *IEEE Transactions on Parallel and Distributed Systems* 4.1, pp. 3–12. DOI: 10.1109/71.205649.
- Alavi, Yousef, Mehdi Behzad, Paul Erdős, and Don R. Lick (1991). “Double Vertex Graphs”. In: *Journal of Combinatorics, Information & System Sciences* 16.1, pp. 37–50.
- Bandelt, H-J and Marcel van De Vel (1989). “Embedding topological median algebras in products of dendrons”. In: *Proceedings of the London Mathematical Society* 3.3, pp. 439–453. DOI: 10.1112/plms/s3-58.3.439.
- Kramer, Florica and Horst Kramer (1969a). “Ein Färbungsproblem der Knotenpunkte eines Graphen bezüglich der Distanz p”. In: *Rev. Roumaine Math. Pures Appl* 14.2, pp. 1031–1038.

References

Kramer, Florica and Horst Kramer (1969b). “Un probleme de coloration des sommets d’un graphe”. In: *CR Acad. Sci. Paris A* 268.7, pp. 46–48.